

Programming Fundamentals - 3st WEEK

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Class : TI-11

JOBSHEET 3 - Operator, Sequence, Flowchart and Pseudocode

2.1 Experiment 1: The Use of Operators

Question:

1. Explain the difference between `x++` and `++x`.
2. What is the result of `int z = x ^ y;`? Please calculate it manually

Documentation:

```
1 package week3.experiment1;
2
3 public class OperatorDewiNovitaSari {
4     Run | Debug
5     public static void main(String[] args) {
6         int x = 10;
7         System.out.println("x++ = " + (x++));
8         System.out.println("After evaluation, x = " + x);
9
10        x = 10;
11        System.out.println("++x = " + (++x));
12        System.out.println("After evaluation, x = " + x);
13
14        int y = 12;
15        System.out.println(x > y || y == x && y <= x);
16
17        int z = x ^ y;
18        System.out.println("The result of x ^ y is " + z);
19
20        z%=2;
21        System.out.println("Final result = " + z);
22    }
23 }
```

Problems 1 Output Debug Console Terminal Ports

```
x++ = 10
After evaluation, x = 11
++x = 11
After evaluation, x = 11
false
The result of x ^ y is 7
Final result = 1
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment1>
```

Answer:

1. The difference between `x++` and `++x`

`x++` means that the current value of x is used first, and then x is incremented by 1.

Whereas `++x` increments x by 1 first, and then the updated value is used.

2. The result of `int z = x ^ y;`

x = 11 = 1011

y = 12 = 1100

1011

1100

____XOR

0111

z = x ^ y = 0111 = 7 So, the result is z = 7.

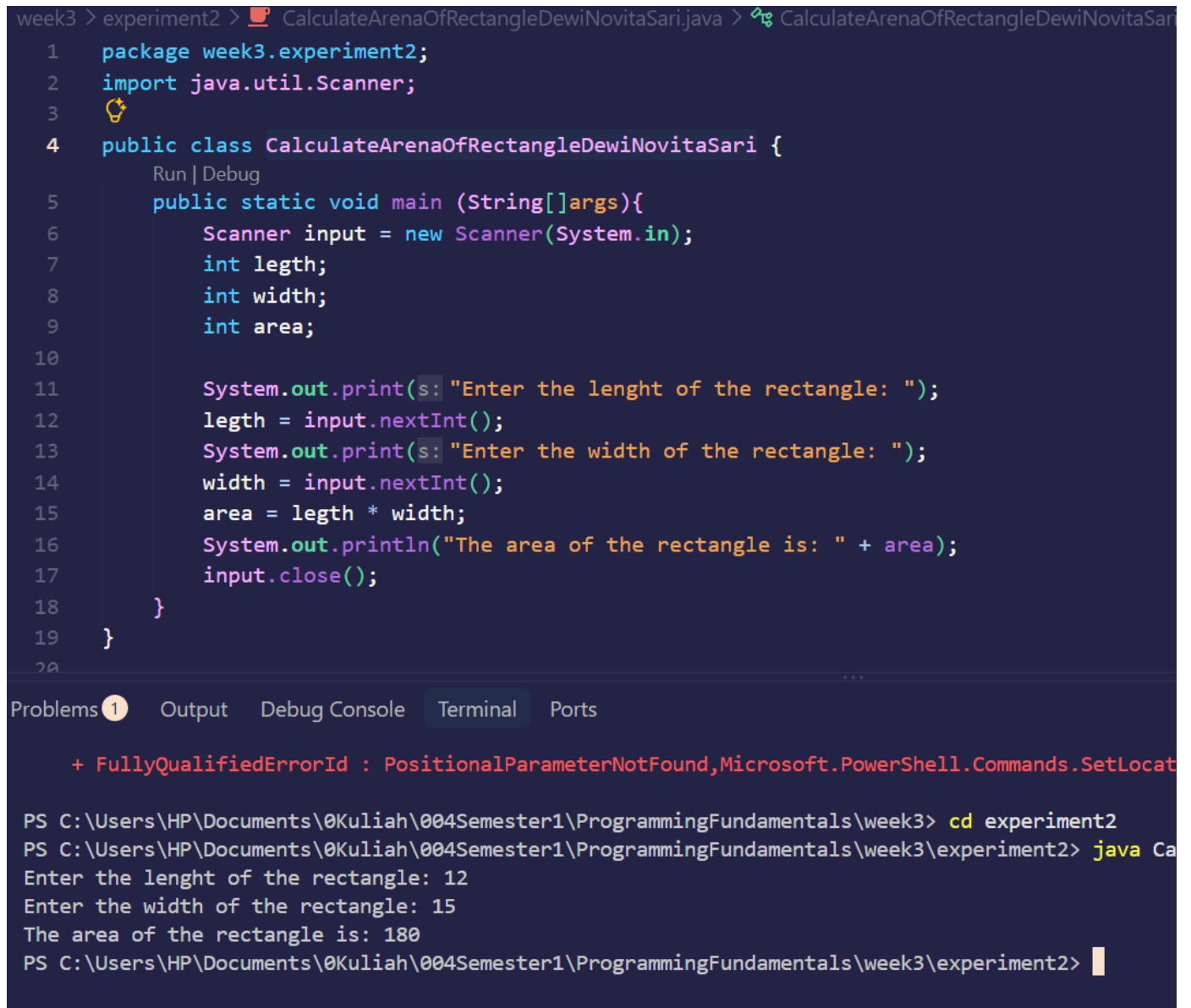
2.2 xperiment 2: Case Study

Question:

1. Explain why it is necessary to declare a Scanner in experiment 2 above!
2. Explain the purpose of the following program snippet.

```
length = input.nextDouble();  
width = input.nextDouble();
```

Documentation:



```
week3 > experiment2 > CalculateArenaOfRectangleDewiNovitaSari.java > CalculateArenaOfRectangleDewiNovitaSari  
1 package week3.experiment2;  
2 import java.util.Scanner;  
3  
4 public class CalculateArenaOfRectangleDewiNovitaSari {  
    Run | Debug  
5     public static void main (String[] args){  
6         Scanner input = new Scanner(System.in);  
7         int legth;  
8         int width;  
9         int area;  
10  
11         System.out.print(s: "Enter the lenght of the rectangle: ");  
12         legth = input.nextInt();  
13         System.out.print(s: "Enter the width of the rectangle: ");  
14         width = input.nextInt();  
15         area = legth * width;  
16         System.out.println("The area of the rectangle is: " + area);  
17         input.close();  
18     }  
19 }  
20
```

Problems 1 Output Debug Console Terminal Ports

```
+ FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.SetLocation  
  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3> cd experiment2  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment2> java Ca  
Enter the lenght of the rectangle: 12  
Enter the width of the rectangle: 15  
The area of the rectangle is: 180  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment2>
```

Answer:

1. We need to declare a `Scanner` because this program needs to accept user input. In this program, the `Scanner` is used to read the length and width of a rectangle entered via the keyboard. In the previous program, the user wasn't asked to enter any data, so there was no need to use a `Scanner`.
2. These statements are used to read integer values entered by the user. The first value is stored in the length variable, while the second value is stored in the width variable. These values are then used to calculate the area of the rectangle.

2.3 Experiment 3: Case Study

Question:

1. Modify the program above by changing the data type of price to double, then run the program and fix any **errors** until it executes correctly and produces the expected output.
2. If, in the program above, the total payment is calculated first and then the discount is computed afterward, will the program still run and produce the expected output? Explain your answer.

Documentation:

1. First

```
CalculateTotalPaymentDewiNovitaSari.java > CalculateTotalPaymentDewiNovitaSari > main(String[])
4 public class CalculateTotalPaymentDewiNovitaSari {
6     public static void main(String[] args){
12         double totalPayment;
13
14         System.out.print(s: "Enter the price of the clothes (Rp:");
15         price = input.nextInt();
16
17         discount = price * DiscountCode;
18         totalPayment = price - discount;
19
20         //Display the result
21         System.out.println("Original Price: Rp"+ price);
22         System.out.println("Discount 15%: Rp"+ discount);
23         System.out.println("Total Payment: Rp"+ totalPayment);
24
25         input.close();
26     }
27
28 }
29
```

Problems Output Debug Console Terminal Ports Run: CalculateTotal

```
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment3> & 'C:\Program Files\Java\jdk-26.0
iew' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\2cf112ee1f5e02
t_ws\experiment3_30ea3a64\bin' 'CalculateTotalPaymentDewiNovitaSari'
Enter the price of the clothes (Rp):45000
Original Price: Rp45000
Discount 15%: Rp6750.0
Total Payment: Rp38250.0
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment3>
```

Answer:

1. Modify the program by changing the data type of price to double

```
5 public class CalculateTotalPaymentDewiNovitaSari {
    Run | Debug
7     public static void main(String[] args){
8         Scanner input = new Scanner(System.in);
9
10        double price;
11        double discount;
12        double DiscountCode = 0.15;
13        double totalPayment;
14
15        System.out.print(s: "Enter the price of the clothes (Rp):");
16        price = input.nextDouble();
17
18        discount = price * DiscountCode;
19        totalPayment = price - discount;
20
21        //Display the result
22        System.out.println("Original Price: Rp"+ price);
23        System.out.println("Discount 15%: Rp"+ discount);
24        System.out.println("Total Payment: Rp"+ totalPayment);
25    }
```

Problems 1 Output Debug Console Terminal Ports

1 error
error: compilation failed
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment3> java Calculate
Enter the price of the clothes (Rp):45.500
Original Price: Rp45.5
Discount 15%: Rp6.825
Total Payment: Rp38.675
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment3>

2. No, the program will not produce the expected output if `totalPayment` is calculated before the discount.

Explanation:

This is because `totalPayment` requires the discount value to calculate the final price. If we calculate `totalPayment` first, Java will have trouble finding the discount value because it hasn't been calculated yet, so it will display an error message because the `discount` variable does not yet have a value.

2.4 Experiment 4: Case Study

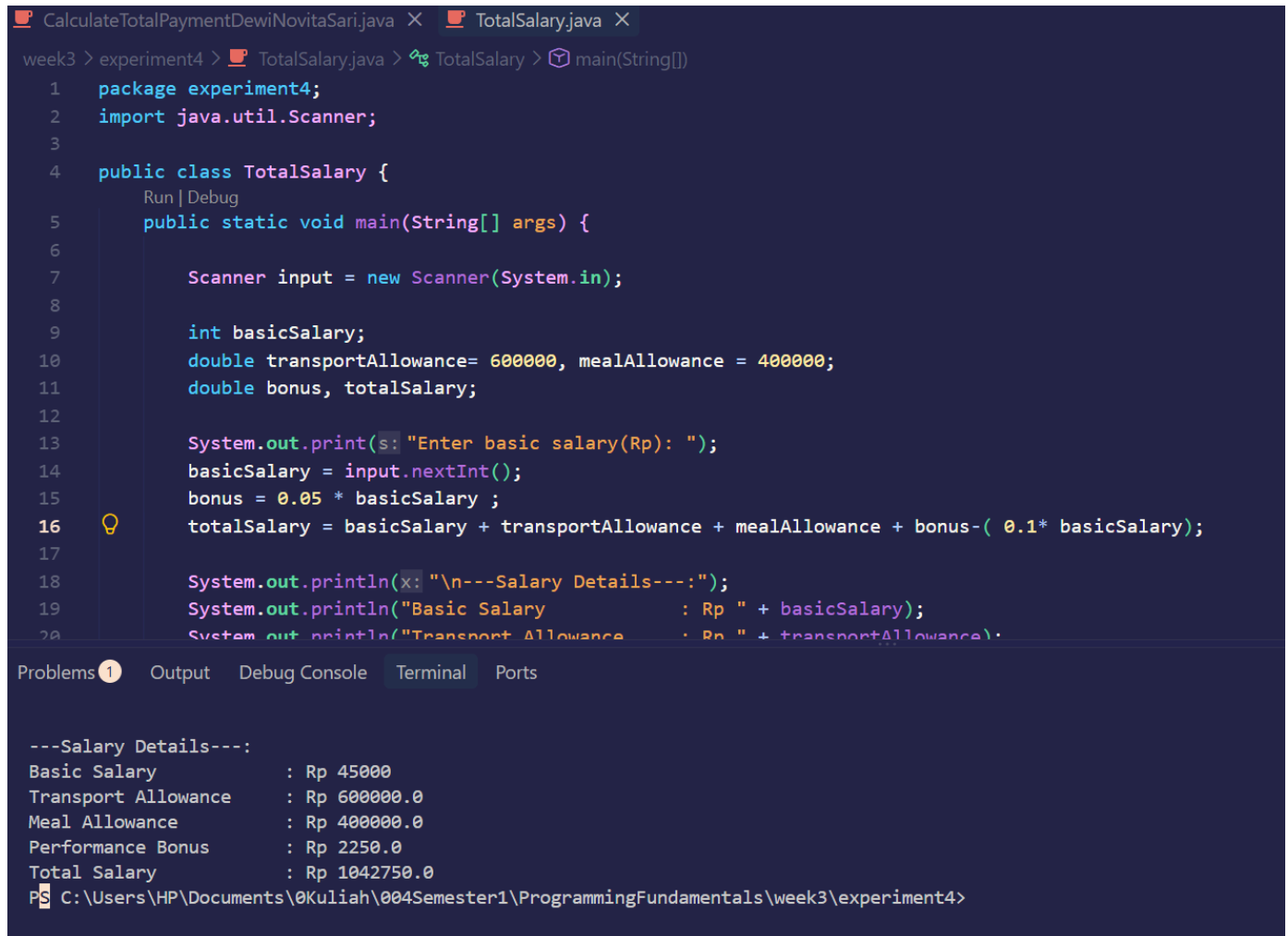
Question:

1. In the following code snippet, what will happen if the parentheses `()` are removed, and does it affect the output result? Explain your answer!

```
totalSalary = basicSalary + transportAllowance + mealAllowance + bonus -  
(0.1* basicSalary);
```

2. Modify the program above so that the total salary output is displayed as an integer value!

Documentation:



The screenshot shows an IDE with two tabs: `CalculateTotalPaymentDewiNovitaSari.java` and `TotalSalary.java`. The active tab is `TotalSalary.java`, showing the following code:

```
1 package experiment4;
2 import java.util.Scanner;
3
4 public class TotalSalary {
5     public static void main(String[] args) {
6
7         Scanner input = new Scanner(System.in);
8
9         int basicSalary;
10        double transportAllowance= 600000, mealAllowance = 400000;
11        double bonus, totalSalary;
12
13        System.out.print(s: "Enter basic salary(Rp): ");
14        basicSalary = input.nextInt();
15        bonus = 0.05 * basicSalary ;
16        totalSalary = basicSalary + transportAllowance + mealAllowance + bonus-( 0.1* basicSalary);
17
18        System.out.println(x: "\n---Salary Details---:");
19        System.out.println("Basic Salary          : Rp " + basicSalary);
20        System.out.println("Transport Allowance : Rp " + transportAllowance);
```

The output window shows the following results:

```
---Salary Details---:
Basic Salary          : Rp 45000
Transport Allowance   : Rp 600000.0
Meal Allowance        : Rp 400000.0
Performance Bonus     : Rp 2250.0
Total Salary          : Rp 1042750.0
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment4>
```

Answer:

1. If the parentheses are removed, the result will remain the same. This is because the multiplication operator `(*)` has higher precedence than addition `(+)` and subtraction `(-)` in the Java programming language. Therefore, `0.1 * basicSalary` will still be evaluated first even without parentheses.

2. Modify the program above so that the total salary output is displayed as an integer value!

```
4 public class TotalSalary {
5     public static void main(String[] args) {
11         double bonus;
12         int totalSalary;
13
14         System.out.print(s: "Enter basic salary(Rp): ");
15         basicSalary = input.nextInt();
16         bonus = 0.05 * basicSalary ;
17         totalSalary =(int)(basicSalary + transportAllowance + mealAllowance + bonus-( 0.1* basicSalary));
18
19         System.out.println(x: "\n---Salary Details---");
20         System.out.println("Basic Salary          : Rp " + basicSalary);
21         System.out.println("Transport Allowance      : Rp " + transportAllowance);
22         System.out.println("Meal Allowance           : Rp " + mealAllowance);
23         System.out.println("Performance Bonus        : Rp " + bonus);
24         System.out.println("Total Salary             : Rp " + (int) totalSalary);
25
26
27         input.close();
28     }
29 }
```

problems 1 Output Debug Console Terminal Ports

```
---Salary Details---
Basic Salary          : Rp 7000000
Transport Allowance    : Rp 600000.0
Meal Allowance         : Rp 400000.0
Performance Bonus      : Rp 350000.0
Total Salary           : Rp 7650000
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\experiment4>
```

Assignments

Taks:

1. Mr Ali buys a motorcycle on credit for Rp. x. He makes a down payment of Rp. y, and the remaining amount is paid in installments over z months with a fixed interest rate of per month on the outstanding balance. Write a program to calculate the monthly installment amount that Mr. Ali must pay.
2. A car travels from Malang to Surabaya, covering x km. The car consumes an average of 1 liter of fuel every 2 km. If the fuel price is Rp. 10,000 per liter, write a program to calculate the total fuel cost required for the trip.

Code:

1. Motorcycle Credits

```
week3 > assignment > MotorcycleCredits.java > MotorcycleCredits > main(String[])
1
2
3 import java.util.Scanner;
4
5 public class MotorcycleCredits {
    Run | Debug
6     public static void main(String[] args) {
7
8         Scanner input = new Scanner(System.in);
9
10        int motorcyclePrice;
11        int downPayment;
12        int monthlyInstallment;
13        int months;
14        double rate = 0.01;
15        int totalmonthlyPayment;
16
17        System.out.print(s: "\n--- Total Monthly Payment Calculation ---\n");
18        System.out.print(s: "Enter the down payment (Rp): ");
19        downPayment = input.nextInt();
20        System.out.print(s: "Enter the installment period (months): ");
21        months = input.nextInt();
22        System.out.print(s: "Enter the motorcycle price (Rp): ");
23        motorcyclePrice = input.nextInt();
24        motorcyclePrice = motorcyclePrice - downPayment;
25        monthlyInstallment = motorcyclePrice / months;
26        rate = (double)motorcyclePrice * rate;
27        totalmonthlyPayment = (monthlyInstallment + (int)rate );
28
29        System.out.println("Total Mounthly Payment: Rp " + totalmonthlyPayment);
30        input.close();
31    }
32 }
```

2. Total Fuel Cost To Trip

```
1 package assignment;
2
3 import java.util.Scanner;
4
5 public class TotalFuelCostToTrip {
    Run | Debug
6     public static void main(String[] args) {
7
8         Scanner input = new Scanner(System.in);
9         // Declare variables
10        double distance;
11        double fuelNeeded;
12        double fuelPrice = 10000;
13        double totalCost;
14        // Input values
15        System.out.print(s: "Enter the distance from Malang to Surabaya (km): ");
16        distance = input.nextDouble();
17        fuelNeeded = distance / 2;
18        totalCost = fuelNeeded * fuelPrice;
19
20        // Display the result
21        System.out.println(x: "\n--- Fuel Cost Calculation ---");
22        System.out.println("Distance : " + distance + " km");
23        System.out.println("Fuel needed : " + fuelNeeded + " liters");
24        System.out.println("Fuel price per liter : Rp " + fuelPrice);
25        System.out.println("Total fuel cost : Rp " + totalCost);
26
27        input.close();
28    }
29 }
```

Output:

1. Motorcycle Credits

```
Problems 1 Output Debug Console Terminal Ports Run: MotorcycleCredits + - [] ...
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\assignment> java MotorcycleCredits.java

--- Total Monthly Payment Calculation ---
Enter the down payment (Rp): 30000000
Enter the installment period (months): 6
Enter the motorcycle price (Rp): 49000000
Total Mounthly Payment: Rp 3356666
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\assignment>
```

2. Total Fuel Cost To Trip

```
Enter the distance from Malang to Surabaya (km): 97
```

```
--- Fuel Cost Calculation ---
```

```
Distance : 97.0 km
```

```
Fuel needed : 48.5 liters
```

```
Fuel price per liter : Rp 10000.0
```

```
Total fuel cost : Rp 485000.0
```

```
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\week3\assignment>
```